Air France Case: Pivot Table and Bubble Chart

This example shows how to generate pivot table to analyze Air France Internet market case. We first read in the data.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('airfrance.csv')
```

```
display(df.info())
display(df.describe().round(2))
display(df.head())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4510 entries, 0 to 4509
Data columns (total 23 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Publisher ID              4510 non-null   object
 1   Publisher Name            4510 non-null   object
 2   Keyword ID                4510 non-null   float64
 3   Keyword                   4510 non-null   object
 4   Match Type                4462 non-null   object
 5   Campaign                  4510 non-null   object
 6   Keyword Group             4510 non-null   object
 7   Category                  4510 non-null   object
 8   Bid Strategy              3286 non-null   object
 9   Keyword Type              4510 non-null   object
 10  Status                    4510 non-null   object
 11  Search Engine Bid         4510 non-null   float64
 12  Clicks                    4510 non-null   float64
 13  Click Charges             4510 non-null   float64
 14  Avg. Cost per Click       4510 non-null   float64
 15  Impressions               4510 non-null   float64
 16  Engine Click Thru %       4510 non-null   float64
 17  Avg. Pos.                 4510 non-null   float64
 18  Trans. Conv. %            4510 non-null   float64
 19  Total Cost/ Trans.        4510 non-null   float64
 20  Amount                    4510 non-null   float64
 21  Total Cost                4510 non-null   float64
 22  Total Volume of Bookings  4510 non-null   float64
dtypes: float64(13), object(10)
memory usage: 810.5+ KB
None
```

	Keyword ID	Search Engine Bid	Clicks	Click Charges	Avg. Cost per Click	Impressions	Engine Click Thru %	Avg. Pos.	Trans. Conv. %	Total Cost/ Trans.	Amount	Total Cost	Total Volume of Bookings
count	4.510000e+03	4510.00	4510.00	4510.00	4510.00	4510.00	4510.00	4510.00	4510.00	4510.00	4510.00	4510.00	4510.00
mean	4.300000e+16	5.43	113.71	167.48	1.89	9283.52	0.11	1.93	0.01	27.61	1033.68	167.48	0.87
std	0.000000e+00	3.30	1062.71	1342.38	1.32	169667.42	0.20	1.08	0.14	226.31	14940.37	1342.38	12.12
min	4.300000e+16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
25%	4.300000e+16	3.39	1.00	2.31	0.83	28.00	0.02	1.14	0.00	0.00	0.00	2.31	0.00
50%	4.300000e+16	6.25	4.00	6.76	1.65	176.00	0.04	1.59	0.00	0.00	0.00	6.76	0.00
75%	4.300000e+16	6.25	19.00	28.49	2.66	843.50	0.11	2.31	0.00	0.00	0.00	28.49	0.00
max	4.300000e+16	27.50	34012.00	46188.44	10.00	8342415.00	2.00	15.00	9.00	9597.17	567463.40	46188.44	439.00

	Publisher ID	Publisher Name	Keyword ID	Keyword	Match Type	Campaign	Keyword Group	Category	Bid Strategy	Keyword Type	...	Click Charges	Avg. Cost per Click	Impressions	Engine Click Thru %	Avg. Pos.	Trans. Conv. %	Total Cost/ Trans.	Amount	Total Cost	To Vol Booki
0	K2615	Yahoo - US	4.300000e+16	fly to florence	Advanced	Western Europe Destinations	Florence	uncategorized	NaN	Unassigned	...	2.31	2.31	11.0	0.09	1.27	9.00	0.26	8777.95	2.31	
1	K2615	Yahoo - US	4.300000e+16	low international airfare	Advanced	Geo Targeted DC	Low International DC	uncategorized	NaN	Unassigned	...	0.63	0.63	6.0	0.17	1.00	1.00	0.63	1574.20	0.63	
2	K2003	MSN - Global	4.300000e+16	air discount france ticket	Broad	Air France Brand & French Destinations	France	uncategorized	Position 2-5 Bid Strategy	Unassigned	...	0.39	0.39	9.0	0.11	1.11	1.00	0.39	390.15	0.39	
3	K1175	Google - Global	4.300000e+16	[airfrance]	Exact	Air France Global Campaign	Air France	airfrance	Position 1-3	Unassigned	...	2.31	0.04	401.0	0.15	2.00	0.03	1.16	1665.15	2.31	
4	K1123	Overture - Global	4.300000e+16	air france online booking	Standard	Unassigned	Unassigned	airfrance	Position 1-2 Target	Unassigned	...	2.20	0.28	318.0	0.03	2.98	0.13	2.20	935.00	2.20	

5 rows × 23 columns

Generate Pivot Table

We use pandas to get a pivot table. The model parameters are:

- **values** - list of fields (coulmns) we want to generate summary stats
- **index** - index for pivot table generated from categorical values of a column. if multiple columns are specified, it will generate a hierarchy of indices. For example using `['Publisher', 'Campaign']`, all campaigns under each publisher will be shown. See example below to add a filter.
- **aggfunc** - method to aggregate, for example sum, mean, ...

```
table = df.pivot_table(values=
                       ['Clicks', 'Click Charges', 'Total Volume of Bookings', 'Amount', 'Impressions'],
                       index=['Publisher Name'], aggfunc="sum", margins=True
                       )
```

```
display(table)
```

Publisher Name	Amount	Click Charges	Clicks	Impressions	Total Volume of Bookings
Google - Global	2365.266667	307.752672	185.483461	4601.338422	2.027990
Google - US	842.820763	170.758740	92.761468	1861.752294	0.748431
MSN - Global	1469.941919	122.832525	113.303030	1413.929293	1.303030
MSN - US	1852.548980	164.270714	110.285714	1735.918367	1.428571
Overture - Global	777.730018	116.267685	110.124774	32366.594937	0.672694
Overture - US	525.617625	214.790076	180.518911	25813.143722	0.437216
Yahoo - US	1389.431417	72.753197	71.807874	1469.834646	1.042520
All	1033.683492	167.476197	113.713747	9283.519734	0.873392

Calcualte derived field for case analysis

`pd.concat()` function joins all dataframes (1 column each) together. axis = 1 for column join.

```
t_f = pd.DataFrame()
```

```
t_f['Avg CPC'] = table['Click Charges']/table['Clicks']
t_f['Prob Booking'] = table['Total Volume of Bookings']/table['Impressions']
t_f['Sum Click Charges'] = table['Click Charges']
```

```
display(t_f)
```

Publisher Name	Avg CPC	Prob Booking	Sum Click Charges
Google - Global	1.659192	0.000441	120946.80
Google - US	1.840837	0.000402	353641.35
MSN - Global	1.084106	0.000922	12160.42
MSN - US	1.489501	0.000823	16098.53
Overture - Global	1.055781	0.000021	64296.03
Overture - US	1.189848	0.000017	141976.24
Yahoo - US	1.013165	0.000709	46198.28
All	1.472788	0.000094	755317.65

Hierarchical index

We want to analyze at campaign level, so use `['Publisher', 'Campaign']`.

```
table2 = df.pivot_table(values=
                        ['Clicks', 'Click Charges', 'Total Volume of Bookings', 'Amount', 'Impressions'],
                        index=['Publisher Name', 'Campaign'], aggfunc="sum", margins=True
                        )
```

```
display(table2)
```

Publisher Name	Campaign	Amount	Click Charges	Clicks	Impressions	Total Volume of Bookings
Google - Global	**Air France Brand & French Destinations**	461567.85	58887.26	35674.0	268252.0	430.0
	Air France Global Campaign	467981.95	62059.54	37221.0	1540074.0	367.0
Google - US	**Air France Branded**	1517842.45	114363.88	106153.0	900619.0	1314.0
	Business Class	144.50	2288.58	443.0	13057.0	1.0
	French Destinations	5103.40	8017.65	3411.0	38080.0	13.0
	Geo Targeted Boston	2878.95	1018.62	391.0	11617.0	4.0
	Geo Targeted Chicago	2740.40	777.30	342.0	5388.0	4.0
	Geo Targeted DC	3300.55	804.04	315.0	6304.0	4.0
	Geo Targeted Detroit	923.95	516.79	186.0	6869.0	1.0
	Geo Targeted Houston	7065.20	1673.87	568.0	20908.0	5.0
	Geo Targeted Los Angeles	2183.65	967.33	371.0	8673.0	2.0
	Geo Targeted Miami	470.05	243.84	112.0	2485.0	1.0
	Geo Targeted New York	33929.45	7926.05	3167.0	51925.0	44.0
	Geo Targeted Philadelphia	434.35	359.85	125.0	1819.0	1.0
	Geo Targeted San Francisco	3822.45	1315.98	526.0	8144.0	3.0
	Geo Targeted Seattle	2817.75	1171.24	542.0	5456.0	4.0
	Google_Yearlong 2006	22373.70	81959.53	30470.0	1803463.0	15.0
	Outside Western Europe	0.00	597.83	216.0	8622.0	0.0
	Paris & France Terms	112370.85	95787.28	29039.0	484619.0	103.0
	Western Europe Destinations	27080.15	33851.69	15732.0	477641.0	31.0
MSN - Global	**Air France Brand & French Destinations**	145524.25	12160.42	11217.0	139979.0	129.0
MSN - US	**Air France Brand & French Destinations**	181549.80	15966.95	10783.0	164928.0	140.0
	Business Class	0.00	128.90	24.0	5191.0	0.0
	Geo Targeted Seattle	0.00	2.68	1.0	1.0	0.0
Overture - Global	**Unassigned**	430084.70	64296.03	60899.0	17898727.0	372.0
Overture - US	**Unassigned**	347433.25	141976.24	119323.0	17062488.0	289.0
Yahoo - US	**Air France Branded**	832028.45	28713.97	37021.0	362691.0	618.0
	Business Class	0.00	851.55	188.0	23630.0	0.0
	French Destinations	1120.30	414.14	290.0	21271.0	1.0
	General Terms	1977.95	606.47	518.0	144298.0	1.0
	Geo Targeted Atlanta	170.00	265.27	129.0	4428.0	1.0
	Geo Targeted Boston	0.00	377.77	147.0	4520.0	0.0
	Geo Targeted Chicago	4403.85	804.50	340.0	7302.0	1.0
	Geo Targeted Cincinnati	0.00	33.76	16.0	676.0	0.0
	Geo Targeted DC	1891.25	467.39	197.0	13571.0	5.0
	Geo Targeted Detroit	0.00	202.29	109.0	2779.0	0.0
	Geo Targeted Houston	0.00	319.01	106.0	4430.0	0.0
	Geo Targeted Los Angeles	0.00	832.77	316.0	9960.0	0.0
	Geo Targeted Miami	0.00	394.69	120.0	4734.0	0.0
	Geo Targeted New York	1650.70	1613.77	572.0	21241.0	2.0
	Geo Targeted Philadelphia	0.00	386.24	139.0	5824.0	0.0
	Geo Targeted San Francisco	0.00	773.49	283.0	17644.0	0.0
	Geo Targeted Seattle	0.00	0.35	1.0	1.0	0.0
	Paris & France Terms	24022.70	6810.04	4201.0	173662.0	19.0
	Western Europe Destinations	15023.75	2330.81	905.0	110683.0	14.0
All		4661912.55	755317.65	512849.0	41868674.0	3939.0

Add a filter

If we want to look at all campaigns under "Google - US", we can just use `table2.loc[index]` since table2 is basically a `DataFrame`.

```
table2.loc['Google - US']
```

Campaign	Amount	Click Charges	Clicks	Impressions	Total Volume of Bookings
Air France Branded	1517842.45	114363.88	106153.0	900619.0	1314.0
Business Class	144.50	2288.58	443.0	13057.0	1.0
French Destinations	5103.40	8017.65	3411.0	38080.0	13.0
Geo Targeted Boston	2878.95	1018.62	391.0	11617.0	4.0
Geo Targeted Chicago	2740.40	777.30	342.0	5388.0	4.0
Geo Targeted DC	3300.55	804.04	315.0	6304.0	4.0
Geo Targeted Detroit	923.95	516.79	186.0	6869.0	1.0
Geo Targeted Houston	7065.20	1673.87	568.0	20908.0	5.0
Geo Targeted Los Angeles	2183.65	967.33	371.0	8673.0	2.0
Geo Targeted Miami	470.05	243.84	112.0	2485.0	1.0
Geo Targeted New York	33929.45	7926.05	3167.0	51925.0	44.0
Geo Targeted Philadelphia	434.35	359.85	125.0	1819.0	1.0
Geo Targeted San Francisco	3822.45	1315.98	526.0	8144.0	3.0
Geo Targeted Seattle	2817.75	1171.24	542.0	5456.0	4.0
Google_Yearlong 2006	22373.70	81959.53	30470.0	1803463.0	15.0
Outside Western Europe	0.00	597.83	216.0	8622.0	0.0
Paris & France Terms	112370.85	95787.28	29039.0	484619.0	103.0
Western Europe Destinations	27080.15	33851.69	15732.0	477641.0	31.0

Bubble Chart

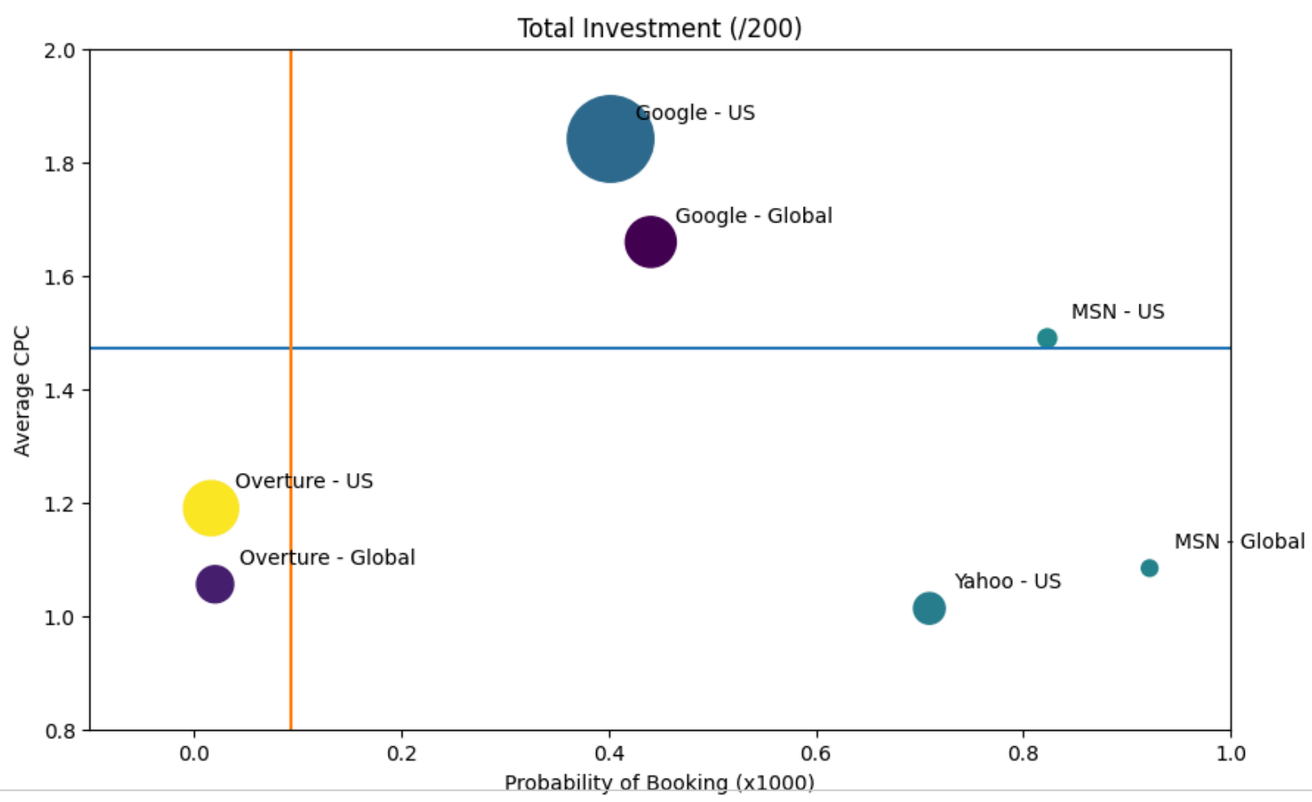
Get all data

- **x** - prob. of booking, rescaled by 1000
- **y** - avg. cost per click
- **bubble size** - total click cost or investment
- **color** - random
- **horizontal quadrant divider** - global avg. cost per click
- **vertical quadrant divider** - prob. of booking

```
x = t_f['Prob Booking'].drop(['All']) * 1000
y = t_f['Avg CPC'].drop(['All'])
size = t_f['Sum Click Charges'].drop(['All']) / 200
color = np.random.rand(7)

x_glb = t_f['Prob Booking']['All'] * 1000
y_glb = t_f['Avg CPC']['All']
```

Plot the chart

1. set figsize 10 x 6 (in 100 pixels)
2. use scatter to plot bubbles
3. plot two dividers
4. annotate publisher names (use t.index)
5. set axis, label, title, etc
6. show and save the plot

```
plt.figure(figsize=(10,6))
plt.xlabel('Probability of Booking (x1000)')
plt.ylabel('Average CPC')
plt.title('Total Investment (/200)')

for xi,yi,i in zip(x, y, table.index):
  plt.annotate(i, (xi,yi), textcoords="offset points", xytext=(12,10))

plt.scatter(x, y, s=size, c=color)
plt.axis([-.1,1,0.8,2])
plt.plot([-.1, 1], [y_glb, y_glb])
plt.plot([x_glb, x_glb], [0.8, 2])


plt.show()
```